

Summer 2025, 4100 Problem Set 4, Hayden Fuller

Please complete all problems. For any proof questions, please structure your proof similar to those from lecture.

1. Use an orthogonal projection to find the point on the plane $S = \{\mathbf{x} \in \mathbb{R}^3 : 3x_1 - x_2 + 4x_3 = 0\}$ that is closest to the point $(-1, 4, 2)$.

$$\text{normal } n = (3, -1, 4)$$

$$\text{point } p = (-1, 4, 2)$$

$$\text{result point } r = p - \frac{p \cdot n}{n \cdot n} n$$

$$p \cdot n = -1 * 3 + 4 * -1 + 2 * 4 = 1$$

$$n \cdot n = 3^2 + (-1)^2 + 4^2 = 26$$

$$r = (-1, 4, 2) - \frac{1}{26}(3, -1, 4) = \left(-\frac{29}{26}, \frac{105}{26}, \frac{48}{26}\right)$$

2. Given vectors, $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$, use Gram-Schmidt to find a corresponding orthonormal set $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3\}$ and show that $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\} = \text{Span}\{\mathbf{q}_1, \mathbf{q}_2\}$.

$$\|\mathbf{v}_1\| = \sqrt{1^2 + 0^2 + 1^2} = \sqrt{2}$$

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\mathbf{v}_2 \cdot \mathbf{q}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

$$\mathbf{u}_2 = \mathbf{v}_2 - (\mathbf{v}_2 \cdot \mathbf{q}_1)\mathbf{q}_1 = \mathbf{v}_2 - \sqrt{2}\mathbf{q}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - \sqrt{2} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

$$\|\mathbf{u}_2\| = \sqrt{1^2 + 1^2 + (-1)^2} = \sqrt{3}$$

$$\mathbf{q}_2 = \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \\ -\frac{\sqrt{3}}{3} \end{bmatrix}$$

$$\mathbf{v}_3 \cdot \mathbf{q}_1 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = -\frac{1}{\sqrt{2}}$$

$$\mathbf{v}_3 \cdot \mathbf{q}_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \cdot \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = -\frac{2}{\sqrt{3}}$$

$$\mathbf{u}_3 = \mathbf{v}_3 - (\mathbf{v}_3 \cdot \mathbf{q}_1)\mathbf{q}_1 - (\mathbf{v}_3 \cdot \mathbf{q}_2)\mathbf{q}_2$$

$$\mathbf{u}_3 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

$$\|\mathbf{u}_3\| = \sqrt{6}$$

$$\mathbf{q}_3 = \begin{bmatrix} -\frac{\sqrt{6}}{6} \\ \frac{2\sqrt{6}}{6} \\ \frac{\sqrt{6}}{6} \end{bmatrix}$$

$$\mathbf{q}_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \mathbf{q}_2 = \begin{bmatrix} \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \\ -\frac{\sqrt{3}}{3} \end{bmatrix}, \mathbf{q}_3 = \begin{bmatrix} -\frac{\sqrt{6}}{6} \\ \frac{2\sqrt{6}}{6} \\ \frac{\sqrt{6}}{6} \end{bmatrix}$$

$$\mathbf{q}_1 = \frac{\mathbf{v}_1}{\sqrt{2}} \in \text{Span}\{\mathbf{v}_1\}, \mathbf{q}_2 = \frac{\mathbf{v}_2 - \mathbf{v}_1}{\sqrt{3}} \in \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$$

$$\mathbf{v}_1 = \sqrt{2}\mathbf{q}_1, \mathbf{v}_2 = \mathbf{v}_1 + (\mathbf{v}_2 - \mathbf{v}_1) = \sqrt{2}\mathbf{q}_1 + \sqrt{3}\mathbf{q}_2$$

$$\text{so } \mathbf{v}_1, \mathbf{v}_2 \in \text{Span}\{\mathbf{q}_1, \mathbf{q}_2\}$$

$$\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\} = \text{Span}\{\mathbf{q}_1, \mathbf{q}_2\}$$

3. Consider the Hilbert space $L^2(\mathbb{R}, w(x))$ where $w(x) = e^{-x^2}$, equipped with the inner product

$$\langle f, g \rangle = \int_{-\infty}^{\infty} f(x)g(x)e^{-x^2}$$

and norm

$$\|f\| = \sqrt{\langle f, f \rangle} = \left(\int_{-\infty}^{\infty} |f(x)|^2 e^{-x^2} \right)^{1/2}.$$

Let $e_n = x^n$ denote the standard basis.

- a. Using the fact that $\int_{-\infty}^{\infty} e^{-x^2} = \sqrt{\pi}$, and using either integration by parts or parameter differentiation, show that $\langle e_i, e_j \rangle = \int_{-\infty}^{\infty} x^{i+j} e^{-x^2} = \begin{cases} \frac{(2n)! \sqrt{\pi}}{2^{2n} n!} & \text{if } i+j = 2n \\ 0 & \text{if } i+j = 2n+1 \end{cases}$

$$I_m = \int_{-\infty}^{\infty} x^m e^{-x^2} dx$$

if m is odd then the integrand is odd, so $I_m = 0$

for m even, integrate by parts

$$I_{2n} = \int_{-\infty}^{\infty} x^{2n} e^{-x^2} dx = \left[-\frac{1}{2} x^{2n-1} e^{-x^2} \right]_{-\infty}^{\infty} + \frac{2n-1}{2} \int_{-\infty}^{\infty} x^{2n-2} e^{-x^2} dx = \frac{2n-1}{2} I_{2n-2}$$

Since $I_0 = \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$, the recursion gives

$$I_{2n} = \frac{(2n-1)(2n-3)\cdots 1}{2^n} I_0 = \frac{(2n)!}{2^{2n} n!} \sqrt{\pi}$$

QED

- b. Use the Gram-Schmidt and the result above to construct an orthonormal set of polynomials from the set $\{1, x, x^2\}$ and use it to find $g \in \mathbb{P}_2$ such that $f(1) =$

$$\int_{-\infty}^{\infty} f(x)g(x)e^{-x^2} \text{ for all } f \in \mathbb{P}_2.$$

$$\langle 1, 1 \rangle = \sqrt{\pi}, \langle 1, x \rangle = 0, \langle 1, x^2 \rangle = \frac{\sqrt{\pi}}{2}$$

$$\langle x, x \rangle = \frac{\sqrt{\pi}}{2}, \langle x, x^2 \rangle = 0, \langle x^2, x^2 \rangle = \frac{3\sqrt{\pi}}{4}$$

$$u_0 = 1$$

$$q_0 = \frac{1}{\|1\|} = \frac{1}{\sqrt{\sqrt{\pi}}} = \pi^{-1/4}$$

since we have $\langle x, 1 \rangle = 0$ we have $u_1 = x$

$$\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{\frac{\sqrt{\pi}}{2}} = \frac{\pi^{1/4}}{\sqrt{2}}$$

$$q_1 = \frac{x}{\|x\|} = \frac{\sqrt{2}}{\pi^{1/4}} x$$

$$\langle x^2, q_0 \rangle = \frac{\sqrt{\pi}/2}{\pi^{1/4}} = \frac{\pi^{1/4}}{2}$$

$$\langle x^2, q_1 \rangle = 0$$

$$u_2 = x^2 - \langle x^2, q_0 \rangle q_0 = x^2 - \frac{1}{2}$$

$$\langle u_2, u_2 \rangle = \frac{3\sqrt{\pi}}{4} - 2 \cdot \frac{1}{2} \cdot \frac{\sqrt{\pi}}{2} + \frac{1}{4} \sqrt{\pi} = \frac{\sqrt{\pi}}{2}$$

$$\|u_2\| = \frac{\pi^{1/4}}{\sqrt{2}}$$

$$q_2 = \frac{u_2}{\|u_2\|} = \frac{\sqrt{2}}{\pi^{1/4}} \left(x^2 - \frac{1}{2} \right)$$

$$q_0 = \pi^{-1/4}, q_1 = \frac{\sqrt{2}}{\pi^{1/4}} x, q_2 = \frac{\sqrt{2}}{\pi^{1/4}} \left(x^2 - \frac{1}{2} \right)$$

$$\begin{aligned}
&g = \sum_{k=0}^2 q_k(1)q_k \\
&\text{because for any } f = \sum \langle f, q_k \rangle 1_k \text{ we get } \langle f, g \rangle = \sum \langle f, q_k \rangle q_k(1) = f(1) \\
&q_0(1) = \pi^{-1/4}, \quad q_1(1) = \frac{\sqrt{2}}{\pi^{1/4}}, \quad q_2(1) = \frac{\sqrt{2}}{2\pi^{1/4}} \\
&g = q_0(1)q_0 + q_1(1)q_1 + q_2(1)q_2 = \pi^{-1/2}(x^2 + 2x + \tfrac{1}{2}) \\
&g(x) = \pi^{-1/2}(x^2 + 2x + \tfrac{1}{2})
\end{aligned}$$

4. Let $\{e_1, \dots, e_m\}$ be an orthonormal subset of $\mathbb{R}^n$. Prove that

$$\|v\|^2 = \langle v, e_1 \rangle^2 + \dots + \langle v, e_m \rangle^2$$

if and only if $v \in \text{Span}(e_1, \dots, e_m)$

Assume $v = \sum_{i=1}^m a_i e_i$ for some scalars a_i .

Because the e_i are orthonormal, $\|v\|^2 = \langle \sum_i a_i e_i, \sum_j a_j e_j \rangle = \sum_{i,j} a_i a_j \langle e_i, e_j \rangle = \sum_i a_i^2$

but $\langle v, e_i \rangle = a_i$, so $\|v\|^2 = \sum_{i=1}^m \langle v, e_i \rangle^2$

conversely assume $\|v\|^2 = \sum_{i=1}^m \langle v, e_i \rangle^2$

define $u = \sum_{i=1}^m \langle v, e_i \rangle e_i$, the orthogonal projection of v onto $\text{Span}(e_1, \dots, e_m)$

$v - u$ is orthogonal to each e_i , hence orthogonal to the whole span

$\|v\|^2 = \|u\|^2 + \|v - u\|^2$, but $\|u\|^2 = \langle \sum_i \langle v, e_i \rangle e_i, \sum_j \langle v, e_j \rangle e_j \rangle = \sum_{i=1}^m \langle v, e_i \rangle^2$

so the assumed equality gives $\|v - u\|^2 = 0$, therefore $v = u \in \text{Span}(e_1, \dots, e_m)$

QED

5. Let $Q \in \mathbb{R}^{n \times n}$ have orthonormal columns. Prove that,

a. The rows of Q must be orthonormal as well

$Q^T Q = I_n$ because the (i, j) entry of $Q^T Q$ is the inner product of col i and col j , which is δ_{ij}

since Q is square and $Q^T Q = I$, Q is invertible with inverse $Q^{-1} = Q^T$, thus $Q Q^T = I_n$

the (i, j) entry of $Q Q^T$ is the dot product of row i of Q with row j of Q .

because $Q Q^T = I$ those dot products equal δ_{ij}

hence distinct rows are orthogonal and each row has length 1, so the rows are orthonormal

QED

b. $\|Qx\| = \|x\|$ for any $x \in \mathbb{R}^n$

$$\|Qx\|^2 = (Qx)^T(Qx) = x^T Q^T Q x = x^T I x = x^T x = \|x\|^2$$

$$\|Qx\| = \|x\|$$

QED

6. Let $\mathbf{u} \in \mathbb{R}^n$ be such that $\|\mathbf{u}\| = 1$ and define the householder matrix, $H = I - 2\mathbf{u}\mathbf{u}^T$.

a. Prove that H is symmetric and that $H = H^{-1}$.

$$H^T = (I - 2\mathbf{u}\mathbf{u}^T)^T = I - 2(\mathbf{u}\mathbf{u}^T)^T = I - 2\mathbf{u}\mathbf{u}^T = H$$

so H is symmetric

$$H^2 = (I - 2\mathbf{u}\mathbf{u}^T)^2 = I - 4\mathbf{u}\mathbf{u}^T + 4\mathbf{u}\mathbf{u}^T\mathbf{u}\mathbf{u}^T$$

$$\mathbf{u}^T\mathbf{u} = 1 \text{ so } \mathbf{u}\mathbf{u}^T\mathbf{u}\mathbf{u}^T = \mathbf{u}(\mathbf{u}^T\mathbf{u})\mathbf{u}^T = \mathbf{u}\mathbf{u}^T \text{ so}$$

$$H^2 = I - 4\mathbf{u}\mathbf{u}^T + 4\mathbf{u}\mathbf{u}^T = I$$

$$\text{so } H^{-1} = H$$

QED

b. Prove that H acts as the identity on the hyperplane, $W = \{\mathbf{v} : \langle \mathbf{u}, \mathbf{v} \rangle = 0\}$

$$\text{if } \mathbf{v} \in W \text{ then } \mathbf{u}^T\mathbf{v} = 0$$

$$\text{hence } H\mathbf{v} = (I - 2\mathbf{u}\mathbf{u}^T)\mathbf{v} = \mathbf{v} - 2\mathbf{u}(\mathbf{u}^T\mathbf{v}) = \mathbf{v} - 2\mathbf{u} \cdot 0 = \mathbf{v}$$

so every vector in W is fixed by H

QED

c. Find a basis $\mathcal{B}$ for $\mathbb{R}^n$ such that $[H]_{\mathcal{B}}$ is diagonal.

pick an orthonormal basis $\{w_1, \dots, w_{n-1}\}$ of the hyperplane W ($\dim(W) = n - 1$)

append $\mathbf{u}$ so $B = \{w_1, \dots, w_{n-1}, \mathbf{u}\}$ is an orthonormal basis of $\mathbb{R}^n$

for each $i \leq n - 1$, $Hw_i = w_i$ by part b, and $H\mathbf{u} = -\mathbf{u}$

therefore the matrix of H in this basis is diagonal with entries $1, \dots, 1, -1$